

PHIL 510: Mathematical Logic and Incompleteness

Term: Fall 2026 | Instructor: Dr. Kurt Gödel (kgodel@philosophy.univ.edu)

Course Description & Overview:

First-order model theory, Peano arithmetic formalizations, Turing computability, and Gödel's incompleteness theorems.

COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

Assignment Description	Weight	Due Module	Deliverable Format
Propositional & First-Order Proof Portfolio	25%	Week 04	Formal Natural Deduction Proofs
Computability & Turing Reduction Problem Set	25%	Week 08	Turing Machine Reduction Proofs
Incompleteness & Formal Systems Term Paper	40%	Week 11	Philosophical & Logical Monograph
Logic Seminar Problem Presentations	10%	Continuous	Board Proof Demonstrations

WEEKLY TERM SCHEDULE & READING REQUIREMENTS

Timeline	Technical Topic Focus	Assigned Readings & Docs
Week 01	Module 01: Syntax & Semantics of First-Order Logic	Enderton (Ch. 1 & 2)
Week 02	Module 02: Model Theory & Tarski Truth	Hodges (Ch. 1–3)
Week 03	Module 03: Completeness Theorem & Compactness	Enderton (Ch. 2.5); Gödel (1929)
Week 04	Module 04: Lowenheim-Skolem & Nonstandard Models	Boolos et al. (Ch. 12 & 13)
Week 05	Module 05: Formal Arithmetic: Peano Axioms	Smullyan (Ch. 1–3)
Week 06	Module 06: Recursive Functions & Computability	Turing (1936); Kleene (Ch. 4)
Week 07	Module 07: Arithmetization & Gödel Numbering	Gödel (1931); Smith (Ch. 5–8)
Week 08	Module 08: The First Incompleteness Theorem	Smith (Ch. 9–14)
Week 09	Module 09: The Second Incompleteness Theorem	Smith (Ch. 15–18)
Week 10	Module 10: Tarski Undefinability of Truth	Tarski (1936); Boolos (Ch. 17)
Week 11	Module 11: Philosophical Implications & Review	Lucas (1961); Penrose (Ch. 2)